
Session 16 (AIML) - Optional Task

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Branch: **Artificial Intelligence and Machine Learning (AIML)**

Domain: **AIML (Artificial Intelligence and Machine Learning)**

Github Repo Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments

Github File Link : https://github.com/vaishnavi-rathiii/mountreach.solutions.internship-aiml-tasks_assignments/blob/main/Session16/IrisEDA%20Optional%20Task.ipynb

Dataset Used: Kaggle - <https://www.kaggle.com/datasets/uciml/iris>

Why I Chose the Iris Dataset ?

- I used the **Iris dataset** for this **Exploratory Data Analysis (EDA)**.
- It is a simple, clean, and widely used dataset in machine learning.
- It contains numerical features and three different flower species.
- It helped me understand the dataset structure and perform data exploration.
- I used it to create visualizations, identify patterns and relationships, and practice basic EDA techniques.
- This analysis helped me understand the data before applying machine learning algorithms.

Program Output:

1. Import Libraries

```
[4]: # Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

2. Load Dataset

```
[24]: # Load the dataset

df = pd.read_csv("Iris.csv")
```

3. First 5 Rows

```
[6]: # Display first 5 rows
print("First 5 Rows:")
df.head()
```

First 5 Rows:

```
[6]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

4. Last 5 Rows

```
[26]: # Display last 5 rows
print("Last 5 Rows:")
df.tail()
```

Last 5 Rows:

```
[26]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

5. Dataset Shape

```
[8]: # Shape of the dataset
print("Shape of Dataset:")
print(df.shape)
```

Shape of Dataset:
(150, 6)

6. Column Names

```
[9]: # Display column names
print("Column Names:")
print(df.columns)
```

Column Names:
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
 'Species'],
 dtype='object')

7. Dataset Information

```
[10]: # Display dataset information  
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 150 entries, 0 to 149  
Data columns (total 6 columns):  
#   Column             Non-Null Count  Dtype    
---  ---               
0   Id                  150 non-null   int64    
1   SepalLengthCm       150 non-null   float64  
2   SepalWidthCm        150 non-null   float64  
3   PetalLengthCm       150 non-null   float64  
4   PetalWidthCm        150 non-null   float64  
5   Species             150 non-null   object   
dtypes: float64(4), int64(1), object(1)  
memory usage: 7.2+ KB
```

8. Data Types

```
[11]: # Display data types  
print(df.dtypes)
```

```
Id                  int64  
SepalLengthCm      float64  
SepalWidthCm       float64  
PetalLengthCm      float64  
PetalWidthCm       float64  
Species            object  
dtype: object
```

9. Summary Statistics

```
[12]: # Display summary statistics  
df.describe()
```

```
[12]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
[12]: # Display summary statistics  
df.describe()
```

```
[12]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

10. Missing Values

```
[13]: # Check missing values  
print(df.isnull().sum())
```

```
Id          0  
SepalLengthCm  0  
SepalWidthCm  0  
PetalLengthCm  0  
PetalWidthCm  0  
Species      0  
dtype: int64
```

11. Unique Values

```
[27]: # Count unique values  
print(df.nunique())
```

```
Id          150  
SepalLengthCm    35  
SepalWidthCm    23  
PetalLengthCm   43  
PetalWidthCm    22  
Species         3  
dtype: int64
```

12. Random Sample

```
[15]: # Display random 5 rows  
df.sample(5)
```

```
[15]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
67	68	5.8	2.7	4.1	1.0	Iris-versicolor
68	69	6.2	2.2	4.5	1.5	Iris-versicolor
17	18	5.1	3.5	1.4	0.3	Iris-setosa
77	78	6.7	3.0	5.0	1.7	Iris-versicolor
114	115	5.8	2.8	5.1	2.4	Iris-virginica

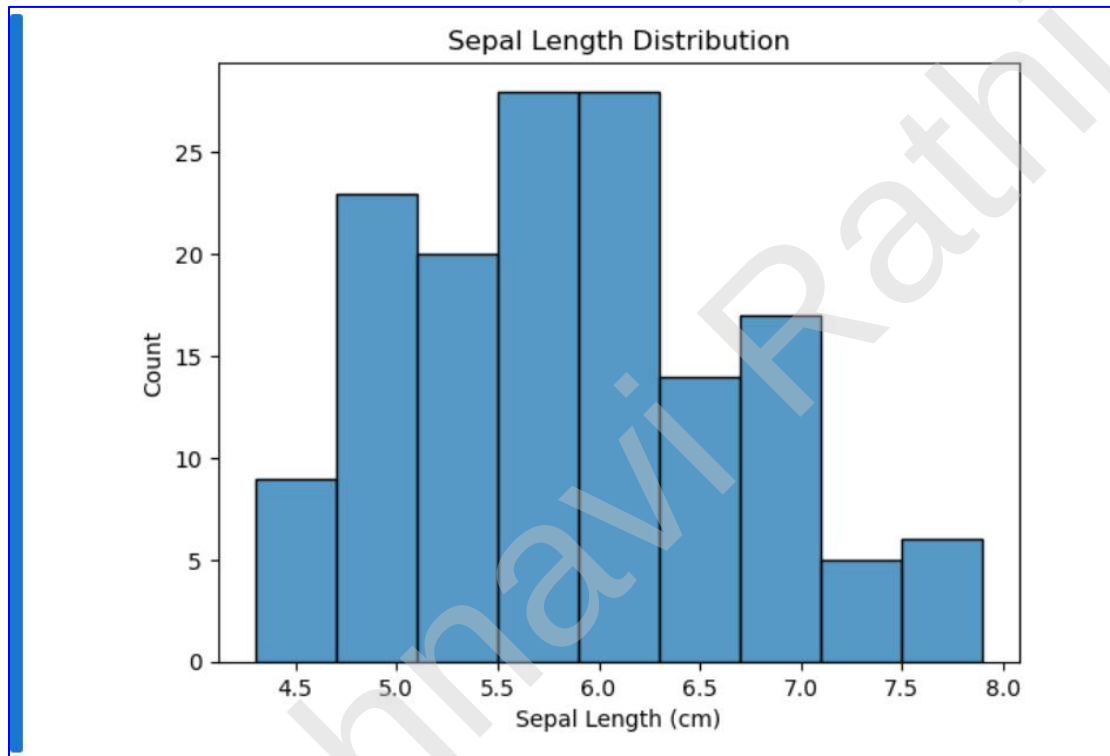
13. Species Count

```
[16]: # Count of each species  
print(df["Species"].value_counts())
```

```
Species  
Iris-setosa      50  
Iris-versicolor  50  
Iris-virginica   50  
Name: count, dtype: int64
```

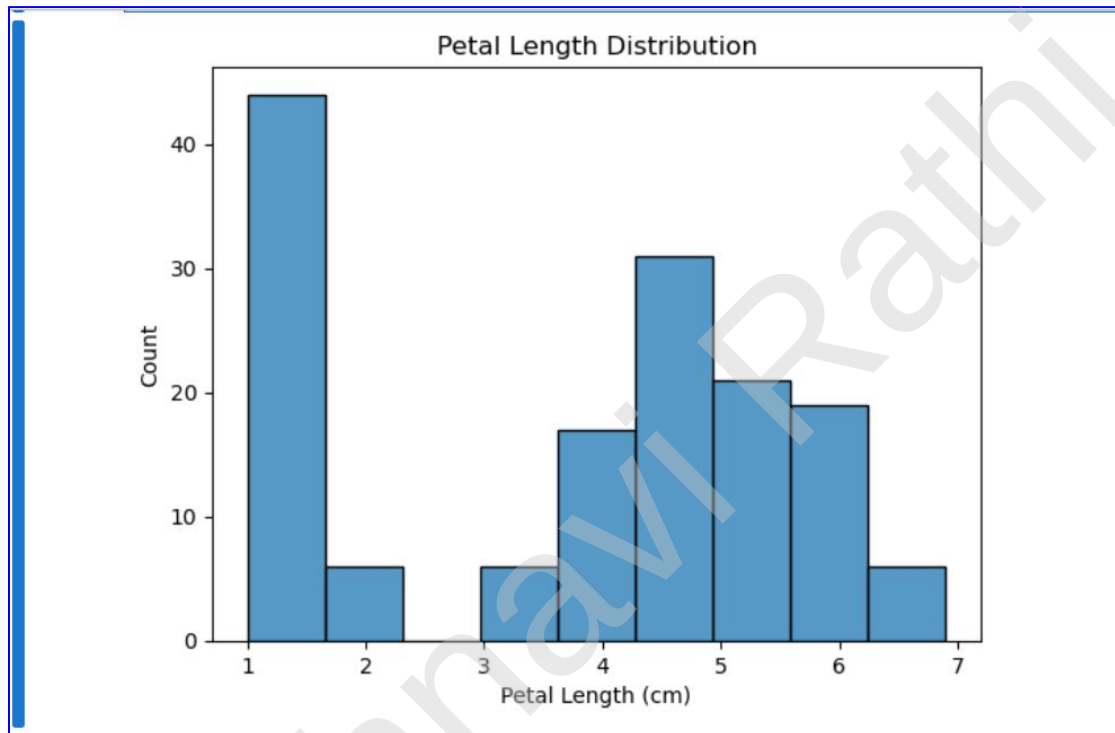
14. Sepal Distribution

```
[17]: sns.histplot(df["SepalLengthCm"])  
  
plt.title("Sepal Length Distribution")  
plt.xlabel("Sepal Length (cm)")  
plt.ylabel("Count")  
  
plt.show()
```



15. Petal Distribution

```
[18]: sns.histplot(df["PetalLengthCm"])  
  
plt.title("Petal Length Distribution")  
plt.xlabel("Petal Length (cm)")  
plt.ylabel("Count")  
  
plt.show()
```

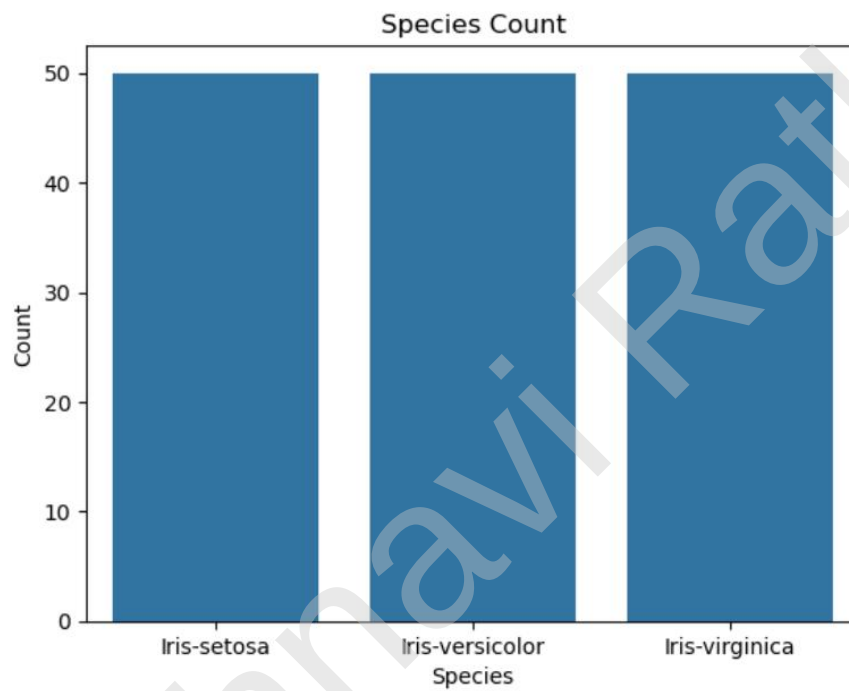


16. Species Count Plot

```
[19]: sns.countplot(x="Species", data=df)

plt.title("Species Count")
plt.xlabel("Species")
plt.ylabel("Count")

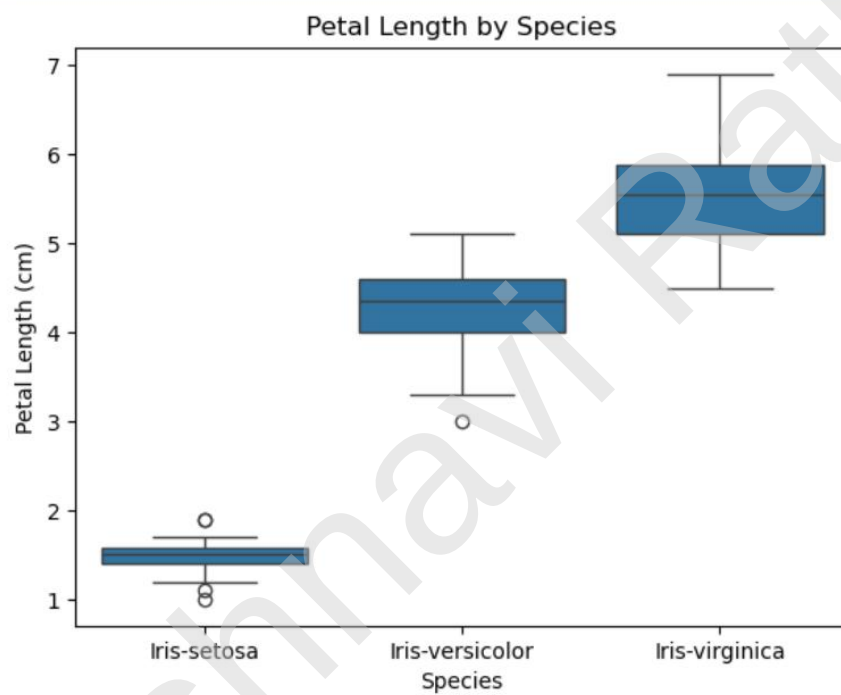
plt.show()
```



17. Species Box Plot

```
[20]: sns.boxplot(x="Species", y="PetalLengthCm", data=df)

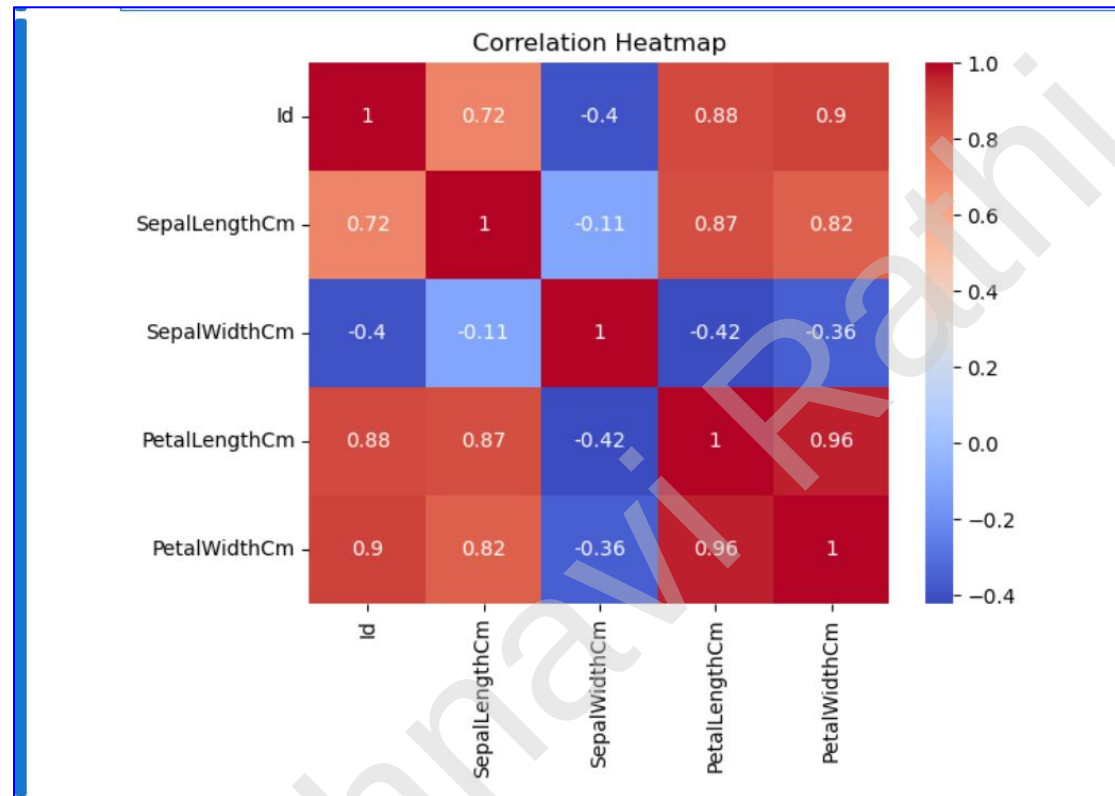
plt.title("Petal Length by Species")
plt.xlabel("Species")
plt.ylabel("Petal Length (cm)")
|
plt.show()
```



18. Correlation Heatmap

```
[22]: corr = df.select_dtypes(include=["number"]).corr()

sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



19. Average Values

```
[23]: print("Average Sepal Length:")
print(df["SepalLengthCm"].mean())

print("\nAverage Petal Length:")
print(df["PetalLengthCm"].mean())
```

Average Sepal Length:
5.843333333333334

Average Petal Length:
3.7586666666666666

20. Analysis Summary

Analysis Summary

- Flower Species
 - The Iris dataset contains three flower species:
 - Iris-setosa
 - Iris-versicolor
 - Iris-virginica
- Data Quality
 - There are no missing values in the dataset.
 - The dataset is clean and suitable for analysis.
- Correlation
 - Petal Length and Petal Width have a strong positive correlation.
- Observations
 - Iris-setosa is clearly distinguishable from the other two species based on petal measurements.
 - Iris-versicolor and Iris-virginica have more similar feature values.
- Conclusion
 - The dataset is clean and suitable for Exploratory Data Analysis (EDA) and machine learning. ""